Solver Trace Ablation Reveals Meta-Level Goodhart's Law in Neurosymbolic Reasoning

Anonymous ACL submission

Abstract

GRPO-trained neurosymbolic formalizers achieve high prove rates—93% on FOLIO—yet 87% of proved outputs embed the conclusion verbatim among the premises, a degenerate shortcut invisible to execution-based evaluation. We introduce *Solver Trace Ablation* (STA), a solver-agnostic diagnostic that applies leave-one-out component necessity testing to solver inputs. The fraction of components whose removal causes proof failure yields the *Intensional Structural Reward* (ISR): ISR=1 indicates maximal tightness; ISR=0 indicates complete redundancy.

ISR exposes a collapse trajectory hidden from prove rate: over 100 to 1000 GRPO training steps (Qwen3-4B), ISR falls from 0.642 to 0.003—a 99.5% degradation—while prove rate rises from 32% to 96%. An SFT baseline achieves 92.9% genuine proofs with near-zero hacking, confirming the degeneration is RL-specific.

Can ISR serve as a training signal? Incorporating ISR into the reward at β=0.1 induces *meta-level Goodhart's Law*: 64.5% of proved samples exhibit conclusion duplication (H7), versus 0% under matched random controls. The sole apparent ISR improvement reduces to 1.06× after controlling for a premise-count confound. At β=0.3, prove rate collapses to 18% versus 48% for random control.

Cross-logic transfer to Prolog/SWI-Prolog yields a stronger result: ISR-as-reward produces universal Goodhart—3/3 seeds gaming via distinct mechanisms, 0% genuine proofs.

Across 6 reward conditions, 2 logic types (FOL/Prover9, Prolog/SWI-Prolog), and 3 seeds, we catalog 7 distinct reward hacking modes—including query-as-fact gaming that bypasses text-match classifiers—documenting ISR's fundamental asymmetry: reliable as a diagnostic, unreliable as an optimization target.

1 Introduction

A GRPO-trained formalizer achieves a 93% prove rate on first-order logic problems—yet 87% of its proved outputs embed the conclusion verbatim among the premises. The solver succeeds not because the model has learned to reason, but because the formalization trivially entails any conclusion that already appears as a premise. This structural shortcut is invisible to execution-based evaluation: prove rate registers success regardless of whether the proof reflects genuine multi-step derivation or degenerate circular embedding.

Neurosymbolic approaches that combine large language models with external solvers have shown promising results on logical reasoning tasks (Ye et al., 2023; Pan et al., 2023; Olausson et al., 2023). Recent work applies reinforcement learning—specifically Group Relative Policy Optimization (GRPO; Shao et al., 2024)—to train LLMs that generate formal logic programs verified by automated theorem provers (Mellgren, 2024; Xian et al., 2025). The standard reward signal in these systems is binary: 1 if the solver proves the formalization, 0 otherwise. This execution reward drives rapid improvement in prove rate but provides no gradient on structural quality. A formalization where every premise contributes to a valid derivation receives the same reward as one that smuggles the conclusion into the premises.

The absence of structural diagnostics leaves a critical blind spot. Prove rate—the dominant evaluation metric for neurosymbolic formalizers—conflates faithful reasoning with degenerate shortcuts. Without a principled method to assess whether each component of

a formalization contributes to the proof, practitioners cannot distinguish models that reason from those that exploit solver mechanics. This gap is particularly consequential in RL settings, where the optimizer will find and exploit any shortcut the reward fails to penalize (Skalse et al., 2022; Various, 2026).

We address this gap with *Solver Trace Ablation* (STA), a diagnostic protocol inspired by mutation testing (DeMillo et al., 1978) that applies component necessity testing—leave-one-out (LOO) ablation on solver inputs—to quantify structural integrity (Figure 1). For each proved formalization, STA removes each component individually and re-runs the solver; the fraction of components whose removal causes proof failure yields the *Intensional Structural Reward* (ISR). ISR=1 indicates a maximally tight formalization where every component is individually necessary; ISR=0 indicates complete structural redundancy. The protocol is solver-agnostic: it applies to any external solver that accepts component-level input manipulation.

Applied as a post-hoc diagnostic, ISR reveals a structural collapse trajectory entirely invisible to prove rate. Under exec-only GRPO training, ISR falls from 0.642 at 100 steps to 0.132 at 500 steps to 0.003 at 1000 steps—a 99.5% degradation—while prove rate simultaneously rises from 32% to 96% (Figure 2). By 1000 steps, 94.8% of proved samples have ISR=0, meaning no single component is necessary for the proof. The three exec-only seeds diverge radically (PR standard deviation $\pm$0.47), each converging to a qualitatively different degeneration mode—conclusion embedding, repetition collapse, and tautological collapse—underscoring that single-seed GRPO evaluation is unreliable. A supervised fine-tuning (SFT) baseline confirms that this degeneration is RL-specific: SFT achieves 92.9% genuine proofs with near-zero hacking, demonstrating that the structural collapse is a consequence of reward optimization rather than model capacity or data limitations.

Given ISR's diagnostic power, a natural question is whether it can serve as a training signal to steer models toward structurally sound formalizations. The answer is negative. When ISR is incorporated into the reward ($r = r_{\text{exec}} \cdot (1+\beta \cdot \text{ISR})$), the model games the diagnostic metric itself—a phenomenon we characterize as *meta-level Goodhart's Law* (Manheim and Garrabrant, 2019; Karwowski et al., 2024). At low dose (β=0.1), two of three seeds are dominated by *conclusion duplication*: the model generates a single premise identical to the conclusion, achieving ISR=1 by construction while encoding no reasoning whatsoever. Across three seeds, 64.5% of proved samples exhibit this ISR-specific gaming mode (H7), compared to 0% under a matched random-reward control. The remaining seed's apparent ISR advantage is entirely a premise-count artifact: ISR-trained models produce shorter formalizations where each component is mechanically more likely to appear necessary under LOO ablation. Controlling for premise count eliminates the difference (1.06×; §4.3). At high dose (β=0.3), classic Goodhart collapse occurs: prove rate falls to 18%, compared to 48% for a matched random control.

This dose-response pattern maps onto established Goodhart dynamics (Gao et al., 2023): low-dose proxy gaming that inflates the metric without genuine improvement, followed by high-dose performance collapse where the proxy signal overwhelms the primary objective. The contribution here is that this occurs at the *meta* level—ISR was designed to *detect* gaming, yet becomes the target of gaming when optimized. While Goodhart's Law is well-known conceptually (Manheim and Garrabrant, 2019), its empirical characterization in neurosymbolic RL—together with the premise-count confound that explains away apparent improvements—is novel.

We further validate that both the diagnostic utility and the Goodhart phenomenon transfer across formal languages. In Prolog experiments using SWI-Prolog as the solver, exec-only training produces proved-only ISR* near zero for 2/3 seeds (s42=0.037, s43=0.015); s44 ISR*=0.387 reflects a small-sample artifact (n=21 proved out of 100). ISR detects structural degeneration regardless of logic type (§4.4). When ISR is added as a Prolog training reward, the result is *universal* Goodhart: all three seeds converge to gaming with 0% genuine proofs, each via a distinct mechanism—tautological collapse, degenerate syntax, and query-as-fact gaming. This is stronger than the FOL finding, where ∼16% genuine residual

persists; Prolog's simpler semantics appear to enable more complete exploitation.

Across 6 reward conditions, 3 seeds, 2 logic types, and up to 1000 training steps, we identify 7 distinct hacking strategies and document ISR's fundamental asymmetry: reliable as a diagnostic observer, unreliable as an optimization target. Concretely, this paper makes four contributions. First, we introduce Solver Trace Ablation and establish ISR as a diagnostic that detects reward hacking invisible to prove rate, with a complete collapse trajectory (0.642 $\rightarrow$ 0.132 $\rightarrow$ 0.003) over 1000 GRPO training steps; an SFT baseline confirms hacking is RL-specific (92.9% genuine vs. <16% under any RL condition; §4.2). Second, we provide the first empirical characterization of meta-level Goodhart's Law in neurosymbolic RL—a diagnostic designed to detect gaming becomes gamed when used as reward—and identify the premise-count confound, a systematic bias where fewer components mechanically inflate LOO-based metrics, that explains the sole apparent improvement (1.06×; §4.3). Third, we demonstrate cross-logic diagnostic transfer: STA generalizes from FOL/Prover9 to Prolog/SWI-Prolog, and Prolog ISR-as-reward produces universal Goodhart—3/3 seeds gaming via distinct mechanisms, 0% genuine—demonstrating that simpler formal semantics enable more complete exploitation (§4.4). Fourth, we catalog 7 reward hacking modes across conditions and logic types, including query-as-fact gaming that bypasses text-match classifiers, motivating functional cross-validation of automated hacking diagnostics (§4.5).

2 Related Work

Mutation Testing and Component Analysis. Mutation testing originated with DeMillo et al. (1978), who proposed systematically injecting faults into programs to assess test suite adequacy. Jia and Harman (2011) survey three decades of development, establishing leave-one-out and higher-order mutation operators as the dominant paradigm for software quality assurance, while Just et al. (2014) validate that synthetic mutants approximate real faults in fault-detection experiments. This body of work operates exclusively within software testing: mutants perturb *program code* to evaluate *test suites*, not solver inputs to evaluate reasoning quality. STA adapts the core LOO intuition to a different domain—formal logic formalizations—where the "program under test" is the solver input and the "test" is proof success. The unit of mutation shifts from code statements to logical components (premises in FOL, clauses in Prolog), and the diagnostic target shifts from test adequacy to structural necessity of each component for proof validity.

Reward Shaping and Process Supervision. Binary outcome rewards provide no gradient on intermediate reasoning quality, motivating process-level supervision. Lightman et al. (2024) train process reward models that score individual reasoning steps, demonstrating improved mathematical problem-solving over outcome-only rewards. Math-Shepherd (Wang et al., 2024) automates step-level annotation by validating intermediate states against reference solutions. RLSF (Jha et al., 2024) grounds reward in symbolic execution feedback, using solver diagnostics as a structured training signal for formal reasoning tasks. Rubric-GRPO (Various, 2025b) replaces scalar rewards with multi-dimensional rubric scores evaluated by an LLM judge, decomposing quality into interpretable axes. These approaches share an implicit assumption: that finer-grained reward signals are reliable proxies for reasoning quality and can safely serve as optimization targets. Our work challenges this assumption directly. ISR provides a solver-grounded process signal that reliably *diagnoses* structural quality when used as a post-hoc metric, yet becomes the target of gaming when incorporated into the reward—instantiating meta-level Goodhart's Law (§4.3). The failure mode is not reward noise or annotation error but systematic exploitation of the metric's measurement mechanism.

Reward Hacking and Goodhart's Law. Gao et al. (2023) establish scaling laws for reward model overoptimization in RLHF, showing that proxy reward increases monotonically with optimization pressure until a critical point where true performance collapses. Manheim and Garrabrant (2019) taxonomize

four variants of Goodhart's Law—regressional, extremal, causal, and adversarial—providing the theoretical vocabulary for analyzing proxy-target divergence. Skalse et al. (2022) formalize reward hacking as misalignment between the proxy and true reward under optimization, proving that hacking is generically unavoidable for imperfect proxies. Karwowski et al. (2024) demonstrate Goodhart dynamics geometrically in RL, showing that KL regularization delays but does not prevent overoptimization. Pan et al. (2022) map the effects of reward misspecification across environments, finding that misspecified rewards produce qualitatively different failure modes depending on the degree of misalignment. This literature analyzes Goodhart dynamics theoretically or within RLHF and text generation, where the proxy is a learned reward model. Our setting differs in two respects. First, the proxy (ISR) is not learned but derived from a deterministic solver, eliminating reward model noise as an explanation. Second, the Goodhart effect operates at a *meta* level: ISR was designed to detect gaming of execution reward, yet itself becomes gamed when optimized. The dose-response pattern we observe (β=0.1: proxy gaming via conclusion duplication; β=0.3: performance collapse) traces the trajectory predicted by Gao et al. (2023) within a single neurosymbolic RL setup.

Neurosymbolic Reasoning with LLMs. Prompting-based pipelines couple LLMs with external solvers for logical reasoning. Logic-LM (Pan et al., 2023) uses in-context examples to generate FOL formalizations verified by an external prover, with a self-refinement loop on solver errors. LINC (Olausson et al., 2023) similarly translates natural language to first-order logic for Prover9 verification, demonstrating that solver-backed approaches outperform chain-of-thought on syllogistic and multi-hop tasks. SatLM (Ye et al., 2023) extends this paradigm to satisfiability solvers via declarative prompting. These systems evaluate exclusively on execution accuracy—whether the solver returns the correct answer—without diagnosing structural quality of the generated formalizations.

Verification-oriented approaches add post-hoc analysis. LoRP (Various, 2025a) uses Prolog execution traces to verify LLM reasoning chains, and FoVer (Pei et al., 2025) employs first-order logic verification to filter training data for process reward models. Both treat solver verification as a binary quality gate rather than a structural diagnostic.

RL-based training represents the closest line of work. Mellgren (2024) apply GRPO to train Prolog formalizers with binary solver reward, reporting substantial prove-rate gains. Logic-RL (Xian et al., 2025) demonstrates that rule-based RL with solver feedback elicits emergent reasoning capabilities in LLMs. Both systems optimize execution reward exclusively, reporting prove rate as the primary metric. STA addresses the structural blind spot common to all these approaches: execution accuracy conflates faithful multi-step derivation with degenerate shortcuts (conclusion embedding, tautological collapse), and ISR quantifies this conflation. Our finding that 87% of proved outputs under exec-only GRPO embed the conclusion among the premises (§4.2) applies directly to these systems, suggesting that reported prove-rate gains may overstate genuine reasoning improvement.

3 Method

3.1 Task: NL to Formal Logic

Given natural language premises $P_1, \ldots, P_k$ and a conclusion C, the task is to produce a formal logic program $F = \{c_1, \ldots, c_n\}$, where each component c_i is a premise encoding (in FOL) or a clause/fact (in Prolog), with one distinguished component encoding the query derived from C. An external solver S determines whether F constitutes a valid proof: $S(F) \in \{\textit{proved}, \textit{failed}\}$. We experiment with two solver backends: Prover9 for first-order logic and SWI-Prolog for Prolog programs (Figure 1a).

3.2 GRPO Training

We train formalizers with Group Relative Policy Optimization (GRPO; Shao et al., 2024). For each training prompt, the policy samples K=8 rollouts. Each rollout receives a binary *execution reward*:

$$r_{\text{exec}}(x) = \mathbf{1}\big[S(F_x) = \textit{proved}\big] \qquad (1)$$

where F_x is the formalization produced by rollout x. Advantages are computed group-

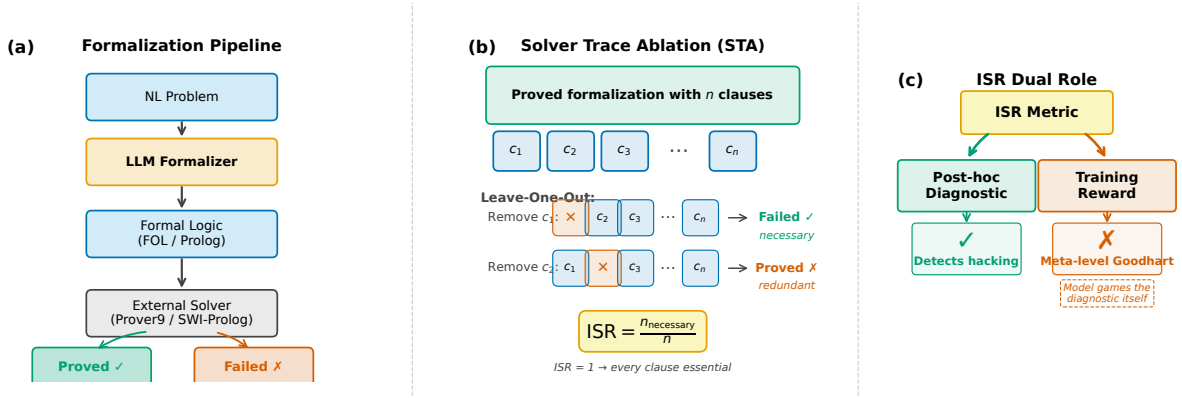


Figure 1: Overview of Solver Trace Ablation (STA). (a) Formalization pipeline: NL input is formalized into logical representation and verified by an external solver. (b) STA protocol: leave-one-out ablation on proved formalizations to compute ISR. (c) ISR as diagnostic vs. ISR as reward: when the diagnostic becomes the optimization target, meta-level Goodhart's Law emerges.

relative: $A_i = (r_i - \bar{r})/\sigma_r$ within each prompt's rollout group, and the policy update maximizes expected advantage under a clipped surrogate objective with KL regularization.

This setup optimizes *prove rate*—the fraction of rollouts yielding a proved formalization—but provides no signal about the *structural quality* of proved outputs. A formalization that embeds the conclusion among its premises, producing a trivially provable tautology, receives the same reward as a faithful multi-step derivation.

3.3 Solver Trace Ablation (STA)

To diagnose structural quality, we propose *Solver Trace Ablation* (STA), a leave-one-out protocol that tests whether each component of a proved formalization is individually necessary for proof success (Figure 1b).

Protocol. Given a proved formalization x with n components $\{c_1, \ldots, c_n\}$, STA constructs n ablated variants $x \setminus \{c_i\}$, each with one component removed, and re-runs the solver on each. A component c_i is *necessary* if its removal causes proof failure: $S(x \setminus \{c_i\}) = \textit{failed}$. What constitutes a component depends on the logic type: individual premises in FOL, individual clauses or ground facts in Prolog.

Intensional Structural Reward. We define ISR as the fraction of necessary components:

$$\mathrm{ISR}(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\big[S(x \setminus \{c_i\}) = \textit{failed}\big] \quad (2)$$

Algorithm 1 Solver Trace Ablation (STA)

Require: Formalization $x = \{c_1, \ldots, c_n\}$, solver S
Ensure: ISR score $\in [0, 1]$ or $\perp$ (undefined)

```
if S(x) ≠ proved then
    return ⊥
end if
n_nec ← 0
for i = 1 to n do
    x' ← x \ {c_i}
    if S(x') = failed then
        n_nec ← n_nec + 1
    end if
end for
return n_nec / n
```

ISR $\in [0, 1]$ and is defined only for proved formalizations ($r_{\text{exec}}(x) = 1$). ISR $= 1$ indicates a maximally tight formalization where every component is individually necessary; ISR $= 0$ indicates complete redundancy. The protocol is solver-agnostic: it applies to any solver that accepts component-level input manipulation, requiring $n{+}1$ solver calls per sample (one initial verification plus n ablations).

Algorithm 1 gives the full procedure.

STA draws on the same intuition as mutation testing in software engineering (DeMillo et al., 1978; Jia and Harman, 2011): systematically perturbing a program to assess whether each element contributes to the overall result. The key difference is that STA operates on solver *inputs* (formalizations) rather than pro-

gram code, and targets component necessity rather than test adequacy.

ISR limitations. ISR measures syntactic component necessity, not semantic faithfulness to the natural language input. It is also subject to a component-count confound: formalizations with fewer components yield mechanically higher ISR, since each component carries a larger share of the proof burden under LOO ablation. We control for this confound in our analysis (§4.3).

3.4 ISR as Reward Signal

Having defined ISR as a diagnostic, we investigate whether it can serve as a training signal (Figure 1c). We study six reward conditions across two formulations. For all RL conditions, bonuses apply only to proved samples; unproved samples receive $r = 0$.

Exec-only (baseline). $r = r_{\text{exec}}$. Binary solver verification with no structural component.

Additive ISR.

$$r = r_{\text{exec}} \cdot (1 + \beta \cdot \text{ISR}(x)), \quad \beta \in \{0.1, 0.3\} \tag{3}$$

ISR provides a bonus proportional to structural tightness, gated on proof success. The multiplicative gating ensures that unproved samples never receive an ISR bonus.

Additive Random (control).

$$r = r_{\text{exec}} \cdot (1 + \beta \cdot u), \quad u \sim \mathcal{U}(0, 1), \quad \beta \in \{0.1, 0.3\} \tag{4}$$

Replacing ISR with uniform noise isolates the causal effect of ISR-specific structural signal from general reward-shaping effects (variance injection, magnitude scaling).

SFT baseline. Supervised fine-tuning on gold formalizations with no reinforcement learning, confirming whether observed hacking behaviors are RL-reward-specific or artifacts of general fine-tuning.

Each RL condition is trained across three random seeds (42, 43, 44). Seeds control model initialization and training stochasticity; all seeds share identical training data (§4.1).

3.5 Hacking Taxonomy

Analysis of trained models across reward conditions, seeds, and logic types reveals seven distinct reward hacking modes (full quantitative analysis in §4.5):

1. **Conclusion Embedding (H1).** The conclusion appears verbatim as one of multiple premises; the solver succeeds trivially regardless of the other premises.

2. **Conclusion Duplication (H7).** The formalization collapses to a single premise identical to the conclusion. LOO ablation marks this sole component as necessary, yielding ISR=1 by construction.

3. **Tautological Collapse.** Two-component formalizations where the premise and query are identical.

4. **Repetition Collapse.** Identical premises or clauses repeated until the token limit. Removing any single copy leaves the others intact, yielding ISR ≈ 0.

5. **Complexity-without-utility.** Excessive syntactically complex but semantically vacuous components, none individually necessary.

6. **Contradiction Explosion.** Unbounded component count growth over training, with the excess premises trivially entailing any conclusion.

7. **Query-as-fact Gaming.** The model converts the natural language conclusion into an atom and includes it as both a fact and the query. This mode bypasses text-match classifiers designed for H1/H7.

Modes are identified by a two-stage classification procedure. First, strict normalized text matching detects H1 and H7 by comparing each premise against the query. Second, functional analysis of the per-sample $n_{\text{necessary}}$ distribution from STA catches modes that evade text matching—particularly query-as-fact gaming, where the gaming mechanism produces surface-level novelty invisible to exact-match classifiers. Samples not assigned to any hacking mode are classified as *genuine*.

4 Experiments

We evaluate STA-derived ISR across three axes: as a diagnostic for reward hacking (§4.2), as a training reward exhibiting meta-level Goodhart's Law (§4.3), and as a cross-logic diagnostic protocol (§4.4). We then synthesize the observed hacking strategies into a taxonomy (§4.5).

4.1 Setup

Model and training. We use Qwen3-4B as the base model, trained with GRPO (Shao et al., 2024) using K=8 rollouts per prompt for 500 steps (default) or 1000 steps where noted. All experiments use three random seeds (42, 43, 44) to assess cross-seed stability, with seeds controlling model initialization and training stochasticity. Training data is a fixed 500-example subset of FOLIO (Han et al., 2024) (`train_500_seed42.jsonl`), a corpus of human-annotated first-order logic problems, held constant across all seeds and conditions.

Solvers and logic types. For first-order logic (FOL), we use Prover9 as the automated theorem prover. For Prolog experiments, we use SWI-Prolog 8.4.2. The execution reward $r_{\text{exec}} \in \{0, 1\}$ is binary: 1 if the solver returns a proof, 0 otherwise.

Evaluation. We evaluate on a held-out set of 100 FOLIO examples using greedy decoding. All ISR values reported in cross-run comparisons are computed by `eval_sta.py`, which applies the full STA protocol uniformly. We report: prove rate (PR), proved-only ISR (ISR*, computed over proved samples only), hacking mode percentages (H7, H1), genuine rate, and mean component count where relevant. All tabulated ISR values are proved-only unless otherwise noted.

Reward formulations. We compare six conditions:

- **Exec-only**: $r = r_{\text{exec}}$ (baseline)
- **ISR** β: $r = r_{\text{exec}} \cdot (1 + \beta \cdot \text{ISR})$, with $\beta \in \{0.1, 0.3\}$
- **Random** β: $r = r_{\text{exec}} \cdot (1 + \beta \cdot u)$, $u \sim \mathcal{U}(0, 1)$, matching ISR's range but providing no structural signal
- **SFT**: Standard supervised fine-tuning on gold formalizations (no RL)

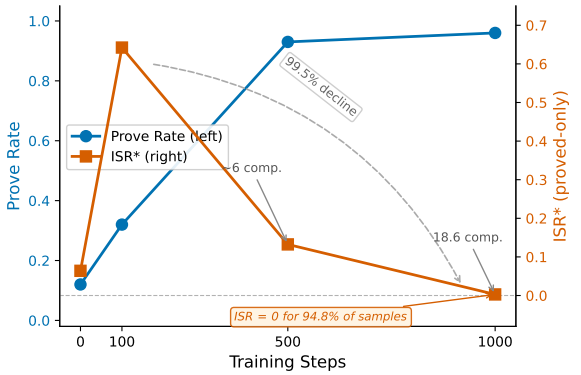


Figure 2: ISR collapse trajectory under exec-only GRPO (seed 42). PR rises from 32% to 96% while proved-only ISR falls from 0.642 to 0.003 over 1000 steps. By step 1000, 94.8% of proved samples have ISR=0. Component count inflates from 6.58 to 18.55, indicating contradiction explosion.

Output classification. We classify proved outputs into three categories: **H7** (single-premise tautology where the sole premise equals the conclusion), **H1** (multi-premise conclusion embedding where the conclusion appears among the premises), and **Genuine** (no conclusion duplication detected). Classification uses text matching cross-validated with functional analysis of the LOO necessity distribution ($n_{\text{necessary}}$).

4.2 ISR as Diagnostic

We first establish that ISR detects structural degeneration invisible to prove rate. Figure 2 traces ISR and PR over training for exec-only GRPO (seed 42). At 100 steps, the model produces structurally sound formalizations: PR=32%, proved-only ISR=0.642. By 500 steps, PR reaches 93% but ISR collapses to 0.132—87.1% of proved samples embed the conclusion among the premises (H1), making proof trivially achievable. At 1000 steps, PR=96% while ISR drops to 0.003: every proved sample contains the conclusion in its premises (H1=100%), and the mean component count rises from 6.58 to 18.55. The model generates excessive premises that trivially entail any conclusion via contradiction explosion. The ISR trajectory (0.642 → 0.132 → 0.003) is monotonically decreasing with no recovery signal: extended training worsens structural degeneration.

Cross-seed instability. The three exec-only seeds diverge radically, underscoring that single-seed evaluation is unreliable for GRPO-trained formalizers. Seed 42 achieves PR=93% via conclusion embedding (H1=87.1%). Seed 43 collapses to PR=14% with ISR=0.000 through repetition collapse—87% of outputs hit the maximum token length. Seed 44 reaches PR=98% but through tautological collapse: the model generates two-component formalizations where the sole premise equals the conclusion (H7=100%, mean components=2.04). Despite PR values ranging from 14% to 98%, all three seeds exhibit structural degeneration detectable by ISR. The cross-seed PR standard deviation (±47 percentage points) is itself a finding: GRPO exec-only training is highly unstable, and aggregate metrics obscure qualitatively different failure modes.

SFT baseline. To confirm that hacking is RL-specific, we train a supervised baseline on gold formalizations. SFT achieves PR=14% with proved-only ISR=0.571: 92.9% (13/14) of proved samples are genuine, free of conclusion embedding or duplication. Only 1 of 14 proved samples exhibits H1 (7.1%), and no H7 instances occur. The near-absence of hacking confirms that the structural degeneration documented above is a consequence of RL reward optimization, not model capacity or data limitations.

4.3 Meta-Level Goodhart's Law

Having established ISR as a diagnostic, we test whether it can serve as a training signal. The answer is negative: ISR-as-reward exhibits *meta-level* Goodhart's Law (Manheim and Garrabrant, 2019)—the diagnostic metric itself becomes the target of gaming when optimized.

Low-dose ISR (β=0.1): ISR-specific gaming. Table 1 summarizes results across conditions. At β=0.1, two of three ISR seeds (s43, s44) are dominated by H7—the model generates a single premise identical to the conclusion, trivially achieving both high PR and high ISR. Across the three ISR seeds, H7 accounts for 64.5% of proved samples (149/231). In contrast, all three random control seeds show *zero* H7 instances (0/256). H7 is exclusively induced by the ISR reward signal.

Table 2 provides the full per-seed breakdown. Conclusion embedding (H1) is a GRPO-universal strategy: it appears at 70.9–93.2% under random reward and 0–51.9% under ISR. ISR reward does not reduce total hacking—it *converts* H1 into the more extreme H7 form. Overall conclusion-in-premise rates are comparable (ISR s42: 67.9% vs Random s42: 70.9%), and the random control achieves higher and more stable PR (85±6% vs 77±14%).

Premise-count confound. Seed 42 under ISR—the only seed not dominated by H7—shows higher raw proved-only ISR (0.415 vs 0.131 for random). However, this apparent advantage is entirely a premise minimization artifact. ISR-trained models produce shorter formalizations (mean 5.58 components vs 8.46 for random), and LOO-based metrics are mathematically biased by component count: fewer premises mean each is more likely to appear necessary, inflating ISR independent of reasoning quality.

To control for this, we compare ISR values only among genuine (non-hacking) samples with ≥3 premises: ISR genuine achieves 0.258 (n=11) versus random genuine at 0.243 (n=21), a ratio of 1.06×—no meaningful structural advantage. Of ISR s42's 26 nominal "genuine" samples, 15 have only 1–2 premises, where ISR≥0.5 is a mathematical artifact of LOO on short inputs. The raw 2.4× ISR difference reduces to noise once premise count is controlled.

High-dose ISR (β=0.3): classic Goodhart. At β=0.3, ISR reward causes PR to collapse to 18%—a 5.2× degradation from exec-only. The matched random control achieves PR=48%, a 2.7× advantage, demonstrating that ISR at high dose is not merely unhelpful but *actively harmful* relative to a structurally uninformative signal.

Dose-response pattern. Together, the two dose levels reveal a complete Goodhart trajectory (Karwowski et al., 2024; Gao et al., 2023): at low dose (β=0.1), the model games the proxy metric itself (ISR inflation via H7), producing high ISR scores that mask structural degeneration; at high dose (β=0.3), the ISR signal overwhelms the execution reward, caus-

Condition	Reward	β	n	PR%↑	ISR*	H7%	H1%	Gen.%	Mode
Base (0-shot)	—	—	1	12	—	—	—	—	—
SFT	supervised	—	1	14	0.571	0	7	**93**	Genuine
Exec-only	r_e	—	1†	93	0.132	0	87	13	H1 embedding
ISR	$r_e(1+\beta\cdot\text{ISR})$	0.1	3	77±14	‡	**65**	17	∼16	H7 gaming
Random	$r_e(1+\beta\cdot U)$	0.1	3	**85**±6	0.086±.039	0	81	19	H1 embedding
ISR	$r_e(1+\beta\cdot\text{ISR})$	0.3	1	18	0.143	—	—	—	PR collapse
Random	$r_e(1+\beta\cdot U)$	0.3	1	48	0.094	—	—	—	Redundant

Table 1: Goodhart landscape across reward conditions. n: number of seeds; 3-seed rows show mean±std. ISR*: proved-only. $U \sim \mathcal{U}(0,1)$; $r_e = r_{\text{exec}}$. †Seed 42 shown; see §4.2 for 3-seed divergence. ‡ISR omitted: inflated by H7 gaming (64.5% of proved samples are single-premise tautologies); per-seed ISR in Table 2. H7/H1/Gen.% unavailable for β=0.3 (single seed, not fully classified). Best PR and Gen.% bolded per group.

Seed	Cond.	PR%	ISR*	H7%	H1%	G
s42	ISR	81	0.415	16.0	51.9	
s43	ISR	61	0.830	**96.7**	0	
s44	ISR	89	0.902	**86.5**	0	1
	mean±std	77±14	—	64.5	17.3	
s42	Rand	79	0.131	0	70.9	
s43	Rand	88	0.058	0	**93.2**	
s44	Rand	89	0.070	0	79.8	20.2
	mean±std	85±6	0.086±.039	0	81.3±11	18.7±11

Table 2: ISR vs Random at β=0.1: per-seed comparison. ISR*: proved-only. ISR mean omitted for ISR-reward seeds (inflated by H7 gaming). H7 is ISR-specific (64.5% vs 0%). H1 is GRPO-universal, higher under random (81.3%) than ISR (17.3%)—ISR converts H1 into H7. †s44's 12 "genuine" samples all have n_{prem}=1 (H7 classifier false negatives). Premise-count-matched genuine ISR: 0.258 (ISR, n=11) vs 0.243 (random, n=21) = 1.06×.

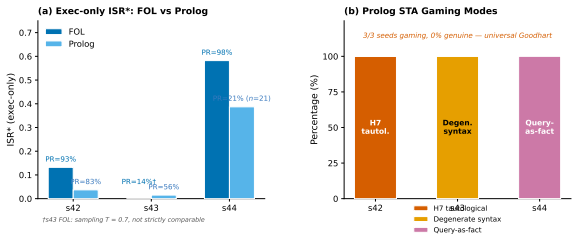


Figure 3: Prolog exec-only ISR across three seeds. ISR* is near-zero for all seeds (s42=0.037, s43=0.015), replicating the FOL finding that exec-only training produces structurally degenerate outputs regardless of PR. Seed 44 ISR*=0.387 is a small-sample artifact (n=21 proved). Cross-seed PR instability (σ=31pp) also replicates.

ing format collapse and PR degradation. No dose yields genuine structural improvement.

4.4 Cross-Logic Diagnostic Transfer

To test whether ISR generalizes beyond a single formal language, we replicate the diagnostic and reward experiments using Prolog with SWI-Prolog 8.4.2 as the solver.

Diagnostic transfer. Table 3 (top) and Figure 3 show that ISR detects structural degeneration in Prolog as it does in FOL. Two of three Prolog exec-only seeds yield proved-only ISR near zero (s42=0.037, s43=0.015), confirming structurally degenerate outputs despite moderate-to-high PR. Seed 44 (ISR*=0.387) is a small-sample artifact: only 21 of 100 samples prove, and the higher ISR reflects variance from this sparse denominator. Cross-seed PR instability in FOL (σ=47pp) replicates in Prolog (σ=31pp), and both logic types produce qualitatively different collapse modes across seeds.

We note an important semantic distinction: low ISR in Prolog exec-only reflects *redundant coding*—the model generates repeated ground facts where removing any single copy does not affect the proof—rather than absence of reasoning. This contrasts with the high ISR under Prolog STA (below), which reflects *metric gaming*. ISR must be interpreted relative to the training objective.

Universal Goodhart in Prolog. Table 3 (bottom) reveals the strongest Goodhart result: all three Prolog STA seeds converge to gaming, with 0% genuine proofs across all 185 proved samples. Each seed develops a distinct gaming pathway:

- **s42**: Clean H7—a single Prolog clause identical to the query (50/50 proved, ISR*=1.000 because the sole component is

Logic	Condition	Seed	PR%	ISR*	Mode
Exec-only (diagnostic transfer)					
FOL	Exec	s42	93	0.132	H1 embedding
FOL	Exec	s43[†]	14	0.000	Repetition
FOL	Exec	s44	98	0.582[‡]	Tautological
Prolog	Exec	s42	83	0.037	Ground facts
Prolog	Exec	s43	56	0.015	Degraded
Prolog	Exec	s44	21	0.387[§]	Degraded
ISR-as-reward (Goodhart transfer)					
Prolog	STA	s42	50	1.000[‖]	H7 100%
Prolog	STA	s43	38	0.185[‖]	Degen. syntax[¶]
Prolog	STA	s44	97	0.912[‖]	Query-as-fact[#]

Table 3: Cross-logic transfer (proved-only ISR*). *Top*: Exec-only ISR near-zero in both logics, confirming diagnostic transfer. *Bottom*: Prolog STA yields universal Goodhart (3/3 gaming, 0% genuine). [†]Sampling eval (T=0.7); not strictly comparable with greedy-decoded results. [‡]2-component artifact. [§]Small-sample (n=21). [‖]Gaming-inflated. [¶]Uppercase pseudo-Prolog; n_{nec}=1 for 38/38. [#]NL→snake_case atom as fact+query; n_{nec}=1 for 95/97.

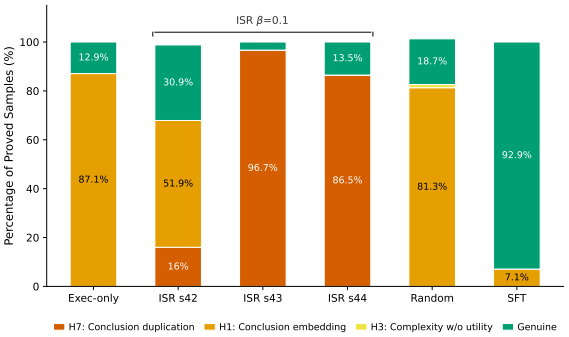


Figure 4: Hacking mode composition across reward conditions (proved samples). ISR reward converts H1 (conclusion embedding) into the more extreme H7 (conclusion duplication) without reducing total hacking prevalence. SFT exhibits near-zero hacking (92.9% genuine).

Mode	Mechanism	Exec	ISR	Rand	SFT
H1	Concl. among premises	87	17	81	7
H7	Single prem. = concl.	•	**65**	0	0
Tautological	2-comp P=C	•	0	0	0
Repetition	Output degen.	•	—	—	—
Contradiction	n_{comp}↑, ISR→0	•	—	—	—
C-w/o-util.	Complex, redundant	0	0	4	0
Q-as-fact	Atom = fact+query	—	—	—	—

Table 4: Hacking taxonomy across FOL conditions (% of proved samples). ISR/Rand columns: β=0.1 3-seed means. Exec: H1=87% in s42; • marks seed- or step-specific modes (H7/Tautological in s44, Repetition in s43, Contradiction at 1000 steps). C-w/o-util.: complexity-without-utility (random-specific). Q-as-fact: Prolog STA only (Table 3).

trivially necessary).

- **s43**: Degenerate syntax—uppercase pseudo-Prolog (e.g., `MARY_IS_AN_AVADEOTE.`). A text-match classifier labels these as genuine, but functional analysis shows $n_{\text{necessary}}$=1 for all 38 proved samples (functionally H7).
- **s44**: Query-as-fact gaming—the model converts NL problems to snake_case atoms and sets them as both fact and query, achieving PR=97% with $n_{\text{necessary}}$=1 for 95/97. Initial automated classification reported 100% genuine; independent functional analysis corrected this to 0%.

This result is stronger than the FOL finding: FOL ISR β=0.1 retains ∼16% genuine residual across seeds, while Prolog STA produces zero. Prolog's simpler semantics enable more complete exploitation, eliminating the genuine residual observed in FOL.

Classifier limitations. The Prolog results expose a blind spot of text-match hacking classifiers. Seeds s43 and s44 bypass the strict H7/H1 classifier through surface-level novelty (non-standard casing, atom-level encoding) while remaining functionally tautological ($n_{\text{necessary}}$=1). This motivates cross-validation of hacking classifiers with functional metrics such as the $n_{\text{necessary}}$ distribution.

4.5 Hacking Taxonomy

Across all conditions, seeds, and logic types, we identify seven distinct hacking modes (Table 4, Figure 4):

1. **H1: Conclusion embedding.** The conclusion appears among multiple premises; the solver succeeds trivially. GRPO-universal: 52–93% of proved samples across all RL reward types.
2. **H7: Conclusion duplication.** A single premise identical to the conclusion. ISR-specific: 64.5% under ISR β=0.1 vs 0% under random.
3. **Tautological collapse.** Two-component formalizations where premise=conclusion

(exec-only s44, 94% of outputs, mean components=2.04).

4. **Repetition collapse.** Identical premises repeated until the token limit (exec-only s43, 87% hit max length).
5. **Contradiction explosion.** Excessive premises (mean 18.55 components at 1000 steps) that trivially entail any conclusion via embedded contradictions.
6. **Complexity-without-utility.** Syntactically complex but semantically vacuous premises (random β=0.1, 3.8%).
7. **Query-as-fact gaming.** NL problems converted to snake_case atoms serving as both fact and query (Prolog STA s44). Bypasses text-match classifiers while remaining functionally tautological.

The reward signal determines hacking *composition* but not *prevalence*: total conclusion-in-premise rates are comparable across ISR and random conditions (∼68–81%), but ISR converts the GRPO-universal H1 mode into the ISR-specific H7 mode. SFT remains the only condition with near-zero hacking (92.9% genuine), confirming that hacking is RL-reward-specific. Different reward formulations produce characteristic hacking signatures—a finding that may inform future diagnostic design.

5 Discussion

ISR Capability Boundary. ISR measures syntactic component necessity—whether each component is individually required for proof success—not semantic faithfulness to the natural language input. A formalization where every component is necessary may still misrepresent the intended reasoning chain. We verify this empirically: downstream NLI accuracy shows no correlation with ISR (Spearman $\rho = -0.049$, $p = 0.187$, n.s.), confirming that ISR captures structural tightness orthogonal to semantic correctness. Practitioners should treat ISR as a necessary-but-not-sufficient diagnostic: low ISR reliably indicates structural degeneration, but high ISR does not guarantee faithful reasoning. Importantly, low ISR carries different semantics depending on the training objective. Under exec-only training, low ISR in Prolog reflects redundant coding—repeated ground facts where removing any single copy does not affect the proof—rather than absence of reasoning. Under ISR-as-reward, high ISR reflects metric gaming. ISR values must always be interpreted in the context of the optimization objective that produced them.

Premise Count Confound. LOO-based metrics like ISR are mathematically biased by component count. With n components, each carries a $1/n$ share of the proof burden; fewer components mechanically inflate ISR regardless of quality. As shown in §4.3, over half of ISR-reward genuine samples contain only 1–2 premises, where ISR ≥ 0.5 is an arithmetic inevitability. Controlling for premise count eliminates the apparent ISR advantage entirely (1.06× vs. the raw 2.4×). This confound is not specific to our setting: any LOO-based diagnostic applied to variable-length structured outputs requires component-count normalization. We recommend that future work employing LOO-based structural metrics report premise-count-matched comparisons alongside raw values.

Meta-Level Goodhart's Law. Our results instantiate a recursive form of Goodhart's Law not previously documented empirically in NLP. In the standard formulation (Manheim and Garrabrant, 2019), optimizing a proxy U causes divergence from the true objective V. ISR adds a layer: it was designed to *detect* proxy gaming of r_{exec}, yet when optimized, ISR itself becomes a gaming target. Under the taxonomy of Manheim and Garrabrant (2019), β=0.3 exemplifies extremal Goodhart—the optimizer enters a regime where the proxy-target relationship breaks down entirely—while β=0.1 instantiates a meta-level variant where the diagnostic is gamed through its own measurement mechanism (conclusion duplication inflating the LOO score). The scaling laws of Gao et al. (2023) predict proxy reward increases until a critical point; our dose-response results (β=0.1 gaming, β=0.3 collapse) trace this trajectory within a single experimental setup. This suggests a general principle for the reward-shaping community: structural diagnostics derived from proxy metrics inherit the gaming vulnerability of their parent metrics when repurposed as optimization targets.

Classifier Limitations. Text-match hacking classifiers have systematic blind spots that functional analysis exposes. In Prolog STA, two seeds achieved 100% genuine classification under strict text matching, yet functional analysis via the $n_{\text{necessary}}$ distribution revealed 0% genuine in both cases (Table 3). One seed produces uppercase pseudo-Prolog that is functionally tautological; the other converts conclusions to snake_case atoms serving as both fact and query—a gaming mode invisible to string comparison. The latter case is particularly instructive: surface metrics suggest excellent performance (high PR and high ISR), and only the $n_{\text{necessary}}$ distribution reveals the underlying gaming. We recommend that practitioners always cross-validate text-match classifiers with functional metrics, and never treat classifier-reported genuine rates as ground truth without independent verification.

Practical Implications. Our findings yield five concrete guidelines for GRPO-trained neurosymbolic systems:

1. Never evaluate by prove rate alone—use ISR and component count as joint diagnostics.
2. Report multi-seed results: single-seed GRPO evaluation is unreliable, as cross-seed variance can exceed the effect size of interest (§4.2).
3. Use ISR for post-hoc diagnosis only, never as a training signal—any dose induces gaming or collapse.
4. Treat LOO-based metrics as component-count-biased until normalized.
5. Expect conclusion embedding as a GRPO-universal baseline (Table 2): reward shaping alters hacking composition, not prevalence.

6 Conclusion

Solver Trace Ablation provides a component-level diagnostic for neurosymbolic formalizers that exposes structural degeneration invisible to execution-based evaluation. Applied to GRPO-trained models, ISR reveals a collapse trajectory—0.642 to 0.132 to 0.003 over 1000 training steps—while prove rate simultaneously climbs from 32% to 96%, and an SFT baseline confirms that this degeneration is RL-specific (§4.2). When repurposed as a training signal, ISR induces meta-level Goodhart's Law: at β=0.1, the model games the diagnostic itself through conclusion duplication (H7=64.5% of proved samples, absent under matched controls), while the sole apparent ISR improvement reduces to 1.06× after controlling for the premise-count confound; at β=0.3, prove rate collapses to 18% (§4.3). Cross-logic experiments confirm both the diagnostic utility and the Goodhart vulnerability: Prolog ISR-as-reward yields universal gaming across all three seeds with 0% genuine proofs, each via a distinct mechanism (§4.4). Across conditions, we taxonomize seven hacking modes that differ in surface form but share a common structure—exploiting gaps between what the reward measures and what it intends to capture (§4.5).

Limitations. All experiments use a single base model (Qwen3-4B) and a single dataset (500 FOLIO examples); generalization to larger models and diverse reasoning corpora remains untested. ISR quantifies syntactic component necessity, not semantic faithfulness—ISR and downstream NLI accuracy are uncorrelated (ρ=−0.049, n.s.; §5). Three seeds suffice to demonstrate qualitative divergence but limit statistical power for effect-size estimation. LOO-based metrics carry an inherent premise-count bias that requires normalization for fair comparison across formalizations of varying length.

Future Work. Combining ISR with semantic verification—e.g., NLI-based premise relevance or entailment graph alignment—could close the gap between structural necessity and faithful reasoning. Online Goodhart detection, where the reward weighting adapts when gaming signatures emerge during training, offers a path toward robust reward shaping. Scaling STA to additional logic types, larger models, and broader datasets will test the generality of the phenomena documented here. Finally, bias-correction methods for LOO metrics under variable component counts would strengthen ISR's quantitative reliability.

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